

# FINITE-INDEX EXTENSIONS OF ESSENTIALLY FREE COUNTABLE BOREL EQUIVALENCE RELATIONS

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ABSTRACT. We show that finite-index extensions of essentially free countable Borel equivalence relations are essentially free, answering a question of Kechris.

A countable Borel equivalence relation is *free* if it is induced by a free Borel action of a countable group. It is *essentially free* if it is Borel bireducible with a free countable Borel equivalence relation, or equivalently if it is Borel reducible to one; see [3, Chapter 10]. For countable Borel equivalence relations  $E \subseteq F$  on the same standard Borel space, we say that  $E$  has *finite index* in  $F$  if each  $F$ -class contains only finitely many  $E$ -classes; no uniform bound is assumed. Kechris asks whether essential freeness is preserved by such extensions [3, Problem 10.4].

**Theorem 1.** *Let  $E \subseteq F$  be countable Borel equivalence relations. If  $E$  is essentially free and has finite index in  $F$ , then  $F$  is essentially free.*

*Proof.* For a standard Borel space  $X$ , let  $\text{Fin}(X)$  be the standard Borel space of its nonempty finite subsets. Given any countable Borel equivalence relation  $R$  on  $X$ , define  $R^{\text{fin}}$  on  $\text{Fin}(X)$  by

$$a R^{\text{fin}} b \iff \{[z]_R : z \in a\} = \{[z]_R : z \in b\}.$$

Now let  $E, F$  be as given. By a result of Kaya [2, Proposition 11], we have  $F \leq_B E^{\text{fin}}$ .

Since  $E$  is essentially free, there exists a free countable Borel equivalence relation  $Q$  on some standard Borel space  $Y$ , and a Borel reduction  $f$  of  $E$  to  $Q$ . The map  $a \mapsto f[a]$  reduces  $E^{\text{fin}}$  to  $Q^{\text{fin}}$ , as also noted by Kaya. It therefore suffices to show that  $Q^{\text{fin}}$  is essentially free.

Let  $\Gamma \curvearrowright Y$  be a free Borel action inducing  $Q$ . For each  $n \geq 1$ , put

$$D_n = \{(y_i)_{i < n} \in Y^n : i \neq j \implies \neg(y_i Q y_j)\}.$$

The wreath product  $\Gamma^n \rtimes \mathfrak{S}_n$  acts on  $D_n$  by

$$((\gamma_i)_{i < n}, \sigma) \cdot (y_i)_{i < n} = (\gamma_i \cdot y_{\sigma^{-1}(i)})_{i < n}.$$

This action is free: if an element fixes a tuple, then  $y_i Q y_{\sigma^{-1}(i)}$  for every  $i$ , so the pairwise inequivalence of the coordinates forces  $\sigma$  to be the identity. Freeness of the original action then gives  $\gamma_i = 1_\Gamma$  for all  $i$ . Let  $R_n$  be the

resulting orbit equivalence relation. Two tuples are  $R_n$ -equivalent exactly when their coordinates represent the same collection of  $Q$ -classes.

Fix a Borel linear order on  $Y$ . For each  $a \in \text{Fin}(Y)$ , choose the least element of  $a$  in each  $Q$ -class that it meets, and list the chosen elements in increasing order. This is a Borel map into the disjoint union of the spaces  $D_n$ , and the preceding description of  $R_n$  shows that it is a reduction:

$$Q^{\text{fin}} \leq_B \bigoplus_{n \geq 1} R_n.$$

Essential freeness is closed under countable disjoint sums and Borel reducibility [3, Chapter 10]. Since each  $R_n$  is free,  $Q^{\text{fin}}$ , and hence  $F$ , is essentially free.  $\square$

Note that the finiteness assumption cannot be replaced by countable index. Indeed, every countable Borel equivalence relation  $F$  on  $X$  extends the equality relation  $\Delta_X$  with countable index, and  $\Delta_X$  is free, being induced by the trivial group.

We conclude by remarking that the present argument does not apply to the corresponding question for treeability is posed in [1, Section 6.4(B)]. The point used above is that the relevant wreath-product actions remain free, but the freeness of the wreath-product actions gives no treeing of their orbit equivalence relations.

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#### REFERENCES

- [1] S. Jackson, A. S. Kechris, and A. Louveau, *Countable Borel equivalence relations*, J. Math. Log. **2** (2002), no. 1, 1–80.
- [2] B. Kaya, *The complexity of the topological conjugacy problem for Toeplitz subshifts*, Israel J. Math. **220** (2017), no. 2, 873–897.
- [3] A. S. Kechris, *The Theory of Countable Borel Equivalence Relations*, Cambridge Tracts in Mathematics, vol. 234, Cambridge University Press, Cambridge, 2024.